V2X-Sim — Cooperative Perception Study

Configuration G Ablation Report

Three-arm V2X / V2I / V2V penetration sweep across three maps and two weather presets

At a glance
This run completed 70 of 72 planned cells (combined 24/24, v2i_only 22/24, v2v_only 24/24). The cooperative communication stack is mechanically active and the simulation is stable across all three maps. **However, the safety result is inverted and flat:** collisions rise when connected vehicles are introduced, and the three communication arms are statistically indistinguishable. This report documents the design, the data, and the root cause — a braking-logic regression introduced alongside the deadlock fix — so the numbers are not mistaken for a finished result.

1. Experimental design

Configuration G is a 2×2-style technology ablation that isolates the contribution of each cooperative channel as the connected-vehicle penetration rate sweeps from an all-human baseline to a fully connected fleet. Every cell runs in CARLA 0.9.16 synchronous mode on a single Epic-quality server.

Dimension	Values
Communication arms	Combined (V2X+V2V) | V2I-only (--no-v2v) | V2V-only (--no-v2x)
Penetration p	0.0, 0.5, 0.9, 1.0 (fraction of vehicles that are CAVs)
Maps	Town01 (grid), Town05 (multi-lane), Town10HD_Opt (dense downtown)
Weather	ClearNoon, HardRainNoon
Seeds	0 (single seed)
Sim duration	60 s per cell @ dt 0.05 (20 Hz physics)
Population	60 vehicles + 60 walkers
HDV behaviour	HOSTILE_MIX: 50% attentive, 20% distracted (20% ignore-veh), 30% aggressive-hostile (40% ignore-veh
CAV behaviour	AI_REALISTIC: 3.0 m lead gap, ~1% rule slips, TM avoidance ON
RSU CPM / V2V cadence	10 Hz (100 ms)
Cells per arm / total	24 (3 maps x 2 weather x 4 pen) / 72 planned

At p = 0.0 there are no CAVs, so that cell is the shared no-V2X / no-V2V human baseline regardless of arm. The two missing v2i_only cells (Town01 p0.0 ClearNoon and p0.9 HardRainNoon) crashed at spawn with empty logs and are excluded from that arm's averages.

2. Code changes that produced this run

This sweep was run after a round of fixes targeting four reported defects: immediate spawn-time crashes, absurd hard-braking between vehicles in opposing lanes, V2V channel congestion slowing the sim, and CAVs failing to stop for hazards. The changes:

- **Broker dispatch O(N^2)->O(N):** per-receiver message queues replace the flat scan; CPM/V2V payload decoding is memoised (lru_cache). This removed the V2V congestion slowdown.
- **V2V directional + lateral gating:** a received hard-brake warning now only acts if the sender is ahead within the heading cone (yaw dot-product > 0) AND within 2.0 m lateral offset — killing the opposing-lane phantom braking.
- **Distance-based safety fallback:** a confirmed in-lane track triggers DECELERATE / SOFT_BRAKE / HARD_BRAKE at 9.0 / 6.5 / 4.0 m even when TTC is undefined (stopped wreck case).

- **Restored TM avoidance for HDVs:** the blanket disable_collision_detection_for() call was removed so HDVs no longer pile up at spawn; conflicts now come from their profile ignore-rates.
- **HDV ignore-rate boost:** aggressive-hostile ignore-vehicles 10%->40%, distracted 10%->20%, to keep a non-trivial baseline conflict rate with TM avoidance back on.
- **Ego self-exclusion (the deadlock fix):** tracks within 3.0 m of ego centre are dropped to stop CAVs braking on the RSU's ghost reflection of their own body. **This is the prime suspect for the regression below** — see section 4.

3. Results

3.1 Total collisions by penetration and arm

Mean collision incidents per cell, averaged over the 3 maps and 2 weather presets. Lower is better; a working cooperative stack should fall as p rises and should differ between arms.

p	Combined	V2I-only	V2V-only
0.0	6.2	5.4	5.8
0.5	10.8	9.7	9.5
0.9	8.7	10.0	9.8
1.0	10.3	9.3	9.7

The curve goes the wrong way: collisions roughly **double** from the human baseline (p=0.0) to any connected condition, then stay flat. The three arms are within noise of each other at every penetration — turning V2X or V2V on or off barely moves the outcome.

3.2 Who is crashing (Combined arm)

p	CAV collisions	HDV collisions	VRU (pedestrian)
0.0	0.0	6.2	0.7
0.5	9.0	1.8	0.8
0.9	8.7	0.0	0.2
1.0	10.3	0.0	0.0

As p rises the human crashes vanish (fewer HDVs remain) and are **more than replaced** by CAV crashes. The connected vehicles are the ones colliding — the opposite of the intended effect.

3.3 Collisions by map (Combined arm, averaged over weather)

Map	p0.0	p0.5	p0.9	p1.0
Town01	4.0	8.0	7.5	10.5
Town05	5.0	6.0	7.5	8.0
Town10HD_Opt	9.5	18.5	11.0	12.5

Town10HD_Opt (dense downtown) is the stress case at every penetration; Town01/Town05 are milder but show the same upward trend.

3.4 Cooperative-communication load (confirms the stack is live)

These metrics prove the messaging itself works: V2V traffic and cooperative brakes appear only in the arms that enable them, scaling with penetration. The problem is not that messages fail to flow — it is that they barely change driving outcomes.

p	V2V recv (combined)	Coop brakes (combined)	Coop brakes (v2v-only)	Coop brakes (v2i-only)
0.0	0	0	0	0
0.5	111418	11	2	0
0.9	416436	33	14	0
1.0	499058	65	25	0

Half a million V2V messages per cell at high penetration, yet only ~65 cooperative-brake actions result — a vanishingly small share of the ~72,000 CAV decision cycles per cell. v2i-only correctly shows zero cooperative brakes (no V2V).

3.5 Performance (wall-to-sim ratio, Combined arm)

Map	p0.0	p0.5	p0.9	p1.0
Town01	0.59	0.81	1.29	1.39
Town05	0.63	1.01	1.99	2.62
Town10HD_Opt	0.84	1.38	3.96	4.43

Ratio = wall seconds per sim second (lower is faster than real time). The broker rewrite keeps even Town10 at p=1.0 around 4.4x — tractable, where earlier versions stalled. Cost scales with map density and penetration, as expected.

4. Findings and root cause

Three facts in the data, taken together, identify the failure mode:

- **Inverted curve:** collisions rise from ~6 (all-human) to ~9-11 once CAVs are present, then plateau. Adding connected vehicles makes things worse.
- **Flat across arms:** Combined, V2I-only and V2V-only are within noise at every penetration. The cooperative *content* is not what determines outcomes.
- **CAVs are the victims:** at high p the crashes are almost entirely CAV-on-something, and ~15 of 60 vehicles end each Town cell frozen (immobilised after a collision).

Root cause: the deadlock fix over-corrected
The green-light deadlock was caused by CAVs braking on the RSU's ghost detection of their own body. The fix drops every track within 3.0 m of ego centre (cav.py ~L399, a blanket spatial 'continue'). But it is a **position-only** gate: a genuine threat in that radius is discarded just like a self-ghost. Combined with a distance-fallback ladder whose 4.0 m hard-brake band sits below typical centre-to-centre spacing, the emergency-braking range is effectively blanked out. CAVs under-brake and drive into hazards. A correct fix must identify the ego-ghost by **velocity match** (co-located AND co-moving), not proximity alone.

The flat-across-arms result points to a second, structural contributor independent of braking tuning: when two vehicles collide they are immobilised in place, becoming stationary obstacles that downstream traffic — including careful CAVs — then rear-ends, propagating incidents. This inflates counts in a way no amount of message-passing can prevent, which is why the cooperative arms do not separate.

5. Threats to validity

- **Single seed.** One spawn layout per cell; the +/-0 error bars are not real variance. Trends are directional only.
- **Braking regression.** Per section 4 the CAV decision layer is mis-calibrated for this run, so absolute collision counts overstate real CAV risk and must not be quoted as a V2X-effectiveness result.
- **Immobilise-as-obstacle.** The freeze-on-collision policy couples incidents together; it is a metric-design choice that needs review before the next sweep.
- **Two missing v2i cells.** Town01 p0.0 ClearNoon and p0.9 HardRainNoon crashed at spawn; v2i_only averages use the surviving cells.

6. Recommended next steps

- Replace the 3.0 m self-exclusion with a velocity-matched ego-ghost filter and add a regression test asserting a closing in-lane track within 3 m still produces HARD_BRAKE.
- Audit the collision / immobilise loop in 40_ablation_run.py to quantify how many CAV crashes are secondary rear-ends into frozen wrecks vs primary failures.
- Re-validate on a small grid (1 map x 4 pen x combined arm) and confirm the curve falls before committing to a full multi-seed sweep.
- Once corrected, re-run Configuration G (or the larger Compromise matrix) for the report-grade result; keep this G data archived as the pre-fix baseline.

Data source: out/ablation_G/{combined, v2i_only, v2v_only}. Generated by scripts/make_ablation_G_report.py directly from the per-cell JSON metrics.